

NUMPY FOR DATA SCIENTISTS

30+ NumPy Methods That Power Real Business Insights

A practical cheat sheet — from everyday basics to advanced simulation — with real business examples for each level.



Easy



Intermediate



Hard

Swipe →

Why NumPy Is the Backbone of Data Science



Speed

Vectorized C loops replace slow Python for-loops — critical for large transaction or clickstream data.



Foundation

pandas, scikit-learn, and TensorFlow are all built on NumPy arrays under the hood.



Business-Ready

Turns raw numbers — sales, ratings, spend — into KPIs, segments, and forecasts in one line.

50x+

faster than pure Python loops on large numeric datasets

Everyday Array Basics

- `np.array()`

Store weekly sales or rating data as one clean structure

- `np.mean()`

Average order value across all transactions

- `np.sum()`

Total quarterly revenue in a single line

- `np.max()` / `np.min()`

Spot the best and worst performing day or SKU

Instant KPIs From Raw Numbers

```
weekly_sales = np.array(  
    [12000, 15500, 9800, 21000])  
  
total = np.sum(weekly_sales)      # 58300  
avg    = np.mean(weekly_sales)    # 14575.0  
best   = np.max(weekly_sales)     # 21000
```

Four numbers in, four business answers out — no manual spreadsheet math.

Filtering & Segmenting Data

- `np.where()`

Label customers or regions "High" vs "Low" value

- `Boolean indexing`

Pull out only the churn-risk customers instantly

- `np.unique()`

Count distinct product categories or store regions

- `np.sort() / argsort()`

Rank your top-performing SKUs by revenue

Segment Without Writing a Loop

```
revenue = np.array([500, 1200, 300, 4800])

segment = np.where(
    revenue > 1000, "High", "Low")

top3 = revenue[np.argsort(revenue)[-3:]]
# array([1200, 4800, ...])
```

One line replaces an entire IF/ELSE block across thousands of rows.

Advanced Analytics & Simulation

- `np.percentile()`

Find the spend level that defines your top 10% (P90) VIPs

- `np.corrcoef()`

Measure how marketing spend correlates with sales

- `np.linalg.solve()`

Optimize budget allocation across channels

- `np.random.normal()`

Run Monte Carlo simulations for revenue risk

Move From Reporting to Forecasting

```
p90 = np.percentile(customer_spend, 90)

corr = np.corrcoef(
    marketing_spend, sales)[0, 1]

np.random.seed(42)
sim = np.random.normal(mu, sigma, 10000)
```

The same tool that computes an average can also simulate next quarter's risk.

10 Easy Methods for Everyday Analysis

`np.array()`

Build arrays from sales, rating, or survey data

`np.zeros()` / `np.ones()`

Initialize placeholder metrics or matrices

`np.arange()`

Generate date or time-step sequences

`np.linspace()`

Create evenly spaced pricing or discount tiers

`np.shape` / `reshape()`

Restructure flat data into rows × columns

`np.sum()`

Total revenue, units sold, or costs

`np.mean()`

Average order value or customer rating

`np.max()` / `np.min()`

Best/worst performing product, day, or region

`np.round()`

Clean currency values for client-ready reports

`np.sqrt()`

Prep step for variance and std-dev calculations

11 Intermediate Methods to Segment & Compare

<code>np.where()</code>	Flag customers/regions as High vs Low value
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<code>Boolean indexing a[a>x]</code>	Filter churn-risk or high-spend customers
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<code>np.unique()</code>	Count distinct categories, regions, or SKUs
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<code>np.sort() / argsort()</code>	Rank top products or reps by performance
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<code>np.std()</code>	Measure volatility in daily or monthly revenue
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<code>np.median()</code>	Robust "typical" value for skewed income data
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<code>np.concatenate/vstack/hstack</code>	Merge datasets from multiple regions
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<code>Broadcasting</code>	Apply a bulk discount % across a price list
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<code>np.cumsum()</code>	Track cumulative revenue growth over time
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<code>np.clip()</code>	Cap outlier discounts or negative inventory values
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<code>np.dot() / @</code>	Compute a weighted score across several KPIs
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10 Advanced Methods for Modeling & Simulation

`np.linalg.solve()`

Optimize budget allocation across channels

`np.linalg.inv() / det()`

Solve multi-variable business equations

`np.percentile()`

Find the VIP customer threshold (e.g. P90 spenders)

`np.corrcoef()`

Correlate marketing spend with sales lift

`np.random.seed/normal()`

Run Monte Carlo simulations for revenue risk

`np.vectorize()`

Apply custom pricing logic across a full dataset

`np.apply_along_axis()`

Run a custom KPI function per row or column

`np.gradient()`

Measure the rate of change in a growth trend

`np.meshgrid()`

Build scenario grids for what-if / sensitivity analysis

`np.einsum()`

Fast weighted aggregation across multiple dimensions

Save This For Your Next Data Project

31 methods across Easy, Intermediate, and Hard — every one grounded in a real business use case.



Easy



Intermediate



Hard

Which method do you use most in your daily work?

Drop it in the comments  — and follow for more Python & Data Science breakdowns.